

Week 1 Module – C & C++ Programming

Topics to be Covered

Students are required to Submit handwritten notes and practice for the following topics in **C and C++ Programming Language**:

1. **Syntax**
2. **Conditions and Loops**
3. **Programming Problems**
 - Solve the given **Star Pattern Questions**

1. Square

```
*****
*****
*****
*****
*****
```

2. Hollow Square

```
*****
*   *
*   *
*   *
*   *
*****
```

3. Triangle

```
*
**
***
****
*****
```

4. Inverted Triangle

```
*****
****
***
**
*
```

5. Pyramid

```
  *
 ***
*****
*****
*****
```

6. Inverted Pyramid

```
*****
*****
*****
***
*
```

7. Half Diamond

```
*
**
***
****
*****
*****
****
***
**
*
```

8. Diamond

```
  *
 ***
*****
*****
*****
*****
***
  *
```

Week-1 Module

[1] ⇒ C Syntax and basic Structure.

C is a procedural and structured programming language. Every standard C program has a precise, structured layout.

- Header Files (`#include <stdio.h>`) ⇒ It's a pre-processor directive that includes the standard I/O library, giving access to `printf()` and `scanf()`.
- Main Function (`int main()`) ⇒ It's the mandatory entry point of a C program. Execution starts at `{` and ends at `return 0;`.
- Output Function (`printf`) ⇒ It's used to display output. It supports format specifiers and escape sequences like `\n`.
- Semicolon (`;`) ⇒ Semicolon terminates every statement in C.

[2] ⇒ Conditions and Loops

- Conditional Statements (`if-else`) ⇒ It runs the `if` block if the condition is true, otherwise runs the optional `else` block.
- Loops (`for loop`) ⇒ It repeats code a fixed number of times using initialization, condition and increment/decrement.

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- Nested Loops $\Rightarrow$ They're needed for 2D patterns.
One loop has to be inside another.

- Outer Loop $\Rightarrow$ It manages the rows
(vertical progression).

- Inner Loop $\Rightarrow$ It handles the columns
(prints spaces or stars horizontally
for the current row).

3 Programming Problems

Q → Solve the given star pattern Questions

1- Square

Output :

```
* * * *
* * * *
* * * *
* * * *
* * * *
```

```
#include <stdio.h>
int main() {
    int n = 5; // Total rows and columns
    // Outer loop for rows
    for (int i = 1; i <= n; i++) {
        // Inner loop for columns
        for (int j = 1; j <= n; j++) {
            printf("*");
        }
        printf("\n"); // Move to the next line
    }
    return 0;
}
```

C Main.c × +

Console

I/O

AI Agent

189 ms

```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          for (int j = 1; j <= n; j++) {
6              printf("*");
7          }
8          printf("\n");
9      }
10     return 0;
11 }
```

```
*****
*****
*****
*****
*****
```

2- Hollow Square Pattern

Output :

```
* * * * *
*       *
*       *
*       *
* * * * *
```

```
#include <stdio.h>
```

```
int main () {
```

```
    int n = 5;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        for (int j = 1; j <= n; j++) {
```

```
            // Print star only at the boundaries
```

```
            if (i == 1 || i == n || j == 1 || j == n) {
```

```
                printf ("*");
```

```
            } else {
```

```
                printf (" "); // Print empty space inside
```

```
            }
```

```
        }
```

```
        printf ("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Main.c

+

Console

I/O

AI Agent

260 ms

```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          for (int j = 1; j <= n; j++) {
6              if (i == 1 || i == n || j == 1 || j == n) {
7                  printf("*");
8              } else {
9                  printf(" ");
10             }
11         }
12         printf("\n");
13     }
14     return 0;
15 }
```

```
*****
*   *
*   *
*   *
*****
```

3- Triangle

Output :

```
  *
 * *
* * *
* * * *
* * * * *
```

```
#include <stdio.h>
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        // Inner loop runs up to the current row number 'i'
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Main.c



```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          for (int j = 1; j <= i; j++) {
6              printf("*");
7          }
8          printf("\n");
9      }
10     return 0;
11 }
```

Console

I/O

AI Agent

185 ms

```
*
**
***
****
*****
```

4- Inverted Triangle

Output :
* * * * *
 * * * *
 * * *
 * *
 *

```
#include <stdio.h>
int main() {
    int n = 5;
    // outer loop counts down from 'n' to 1
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Main.c



```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = n; i >= 1; i--) {
5          for (int j = 1; j <= i; j++) {
6              printf("*");
7          }
8          printf("\n");
9      }
10     return 0;
11 }
```

Console

I/O

AI Agent

213 ms

```
*****
*****
***
**
*
```



5- Pyramid

Output :

```
  *
 * * *
 * * * * *
 * * * * * *
 * * * * * * *
```

```
#include <stdio.h>
int main () {
    int n=5;
    for (int i=1; i<=n; i++) {
        // Print leading spaces
        for (int j=1; j<=n-i; j++) {
            printf(" ");
        }
        // Print stars based on odd number formula
        for (int k=1; k<=(2*i-1); k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

C Main.c × +

```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          for (int j = 1; j <= n - i; j++) {
6              printf(" ");
7          }
8          for (int k = 1; k <= (2 * i - 1); k++) {
9              printf("*");
10         }
11         printf("\n");
12     }
13     return 0;
14 }
```

Console

I/O

AI Agent

198 ms

```
*
***
*****
*****
*****
```

6- Inverted Pyramid

Output :

```
* * * * *
 * * * * 
  * * * * 
   * * * 
    * * 
     *
```

```
#include <stdio.h>
int main () {
    int n = 5;
    // Outer loop handles rows in reverse order
    for (int i = n; i >= 1; i--) {
        // Print spaces
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // Print stars
        for (int k = 1; k <= (2 * i - 1); k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

C Main.c × +

```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = n; i >= 1; i--) {
5          for (int j = 1; j <= n - i; j++) {
6              printf(" ");
7          }
8          for (int k = 1; k <= (2 * i - 1); k++) {
9              printf("*");
10         }
11         printf("\n");
12     }
13     return 0;
14 }
```

Console

I/O

AI Agent

190 ms

*

7- Half Diamond

Output: *

```
  * *
 * * *
* * * *
* * * * *
 * * * *
  * * *
   * *
    *
```

```
#include <stdio.h>
int main () {
    int n = 5;
    // Upper half : Growing triangle
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf ("*");
        }
        printf ("\n");
    }
    // Lower half : Shrinking inverted triangle
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            printf ("*");
        }
        printf ("\n");
    }
    return 0;
}
```

+

212 ms 

*
 **

 **
 *

8- Diamond

Output :

```
  *
 ***
*****
*****
*****
 ***
  *
```

```
#include <stdio.h>
int main() {
    int n = 5;
    // Upper half : Regular Pyramid
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n - i; j++) printf(" ");
        for (int k = 1; k <= (2 * i - 1); k++) printf("*");
        printf("\n");
    }
    // Lower half : Inverted Pyramid
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= n - i; j++) printf(" ");
        for (int k = 1; k <= (2 * i - 1); k++) printf("*");
        printf("\n");
    }
    return 0;
}
```

C Main.c X +

+

Console

☐ I/O

AI Agent

194 ms 

```
1  #include <stdio.h>
2  int main() {
3      int n = 5;
4      for (int i = 1; i <= n; i++) {
5          for (int j = 1; j <= n - i; j++) printf(" ");
6          for (int k = 1; k <= (2 * i - 1); k++) printf("*");
7          printf("\n");
8      }
9      for (int i = n - 1; i >= 1; i--) {
10         for (int j = 1; j <= n - i; j++) printf(" ");
11         for (int k = 1; k <= (2 * i - 1); k++) printf("*");
12         printf("\n");
13     }
14     return 0;
15 }
```

*

*